Modern Crypto

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Homework 1

Q1. (a) OTP scheme where length of key is 100 times size of bits in message, n. first 99n bits are revealed of key and last n bits are hidden to be used for OTP of n-bit message. Hence scheme still achieves perfect secrecy even with 99% of key revealed.

In perfect secrecy, $\forall c \in C, \forall m \in M, \exists m, m' \in M$ s.t. $m \neq m'$

$$Pr[K \leftarrow Gen : Enc_K(m) = c] = Pr[K \leftarrow Gen : Enc_K(m') = c]$$

Proof:

let n = bits of message length

$l > n$ where $l = 100n$

$K = \{k_{reveal}, k_{secret}\}$ where $k_{reveal} \in \{0,1\}^{0.99l}$, $k_{secret} \in \{0,1\}^{0.01l}$

Encrypt message w/ OTP as $Enc(k_{secret}, m) \rightarrow c$

$k_{secret} = k_1, k_2, \ldots, k_n$

$$c' = m \oplus (k_1, k_2, \ldots, k_n)$$

$$c = (k_{reveal}, c') \quad \leftarrow \text{reveal first 99\% of key}$$

Decrypt message $Dec(k_{secret}, c) \rightarrow m$

$c = (k_{reveal}, c')$

$m = c' \oplus (k_1, k_2, \ldots, k_n)$ using k_{secret}

(b) Use key K of size λ in OTP scheme twice for messages m_1 and m_2 with length λ each. This is not perfectly secure because re-using the same xor-key for 2 messages reveals their xor difference to a PPT adversary. Furthermore the probability remains $2^{-\lambda}$ since recovering which key was used the ciphertext does not increase chances of finding K.

Proof: $m_1, m_2 \in \{0,1\}^\lambda$

Gen $K = \{0,1\}^\lambda$ uniformly at random

Enc (m_1, m_2)

$$c = (c_1, c_2) = (m_1 \oplus k, m_2 \oplus k)$$

$Dec_K(c_1, c_2)$:

$$m_1 = c_1 \oplus k, \quad m_2 = c_2 \oplus k$$

From scheme, $c_1 \oplus c_2 = (m_1 \oplus k) \oplus (m_2 \oplus k) = m_1 \oplus m_2$.
So message xor (difference) can be found by A but for 2^λ keys, each is only consistent with same pair of messages so $Pr[\text{"}k_{adversary} = K] \leq 2^{-\lambda}$

Q2 $f: \mathbb{N} \to \mathbb{R}$ is negl(n) if $\forall c, \exists N$ s.t. $\forall n > N$
$f(n) < n^{-c}$

or $\exists N$ s.t. $\forall n > N$, $f(n) < 1/p(n)$

① $f(n) = 1/n^{100}$

if $c = 101$ then $f(n) < n^{-101}$ where $n^{-100} > n^{-101}$
so n^{-100} is not negligible since $\exists c$ where $f(n) \not< n^{-c}$

② $f(n) = 2^{-100 \log n}$

$2^{-100 \log n} \Rightarrow (2^{\log n})^{-100} \Rightarrow n^{-100}$ (same as Q2.1)

Like, the previous problem, $2^{-100 \log n}$ is <u>non negligible</u>

③ $f(n) = n^{-\log n}$

$\forall c, \exists N$ s.t. $\forall n > N, f(n) < n^{-c}$

since $\frac{1}{n^{\log n}} < \frac{1}{n^c}$ as n grows large, $f(n) = n^{-\log n}$

<u>is negligible</u> as it decreases faster than any power $1/n^c$

④ $f(n) = \text{poly}(n) \cdot \text{negl}(n)$

here, $\text{poly}(n) = n^d$ and $\text{negl}(n) = \frac{1}{n^{c+d}}$
where d is the degree of $\text{poly}(n)$

so $\text{poly}(n) \cdot \text{negl}(n) \leq n^d \cdot \frac{1}{n^{c+d}} \leq \frac{1}{n^c}$ which meets the requirements for $\text{negl} f: \mathbb{N} \to \mathbb{R}$. hence $\text{poly}(n) \cdot \text{negl}(n)$ is <u>negligible</u>

Q3. Let $p_0 = \Pr_{x \leftarrow D_0}[A(x) = 1]$

$p_1 = \Pr_{x \leftarrow D_1}[A(x) = 1]$

since $\left| \Pr_{x \leftarrow D_0}[A(x) = 1] - \Pr_{x \leftarrow D_1}[A(x) = 1] \right| = \varepsilon$

then $|p_0 - p_1| = \varepsilon$ $\leftarrow p_1 > p_0$?

A guesses b as $b' = \begin{cases} 0 \text{ if } A(x) = 1 & D_0 \\ 1 \text{ if } A(x) = 0 & D_1 \end{cases}$

$\Pr_{b \leftarrow \{0,1\}, x \leftarrow D_b}[b' = b] = \frac{1}{2} p_0 + \frac{1}{2}(1 - p_1) = \frac{1}{2} + \frac{1}{2}(p_0 - p_1)$

$\Rightarrow \frac{1}{2} + \frac{1}{2}\varepsilon$

Distinguishing D_0 & D_1 with ε $\iff$ Guessing hidden bit b with $\frac{1}{2} + \frac{\varepsilon}{2}$

Q4. (1) An efficient alg. to check $a \in \mathbb{Z}_p^*$ is a quadratic residue for odd prime p, we can use Euler's Criterion

$$\left(\frac{a}{p}\right) = a^{\frac{p-1}{2}} \bmod p = \begin{cases} 1 \pmod p & \text{if } a \text{ is quadratic residue mod } p \\ -1 \pmod p & \text{if } a \text{ is non residue mod } p \end{cases}$$

To check if a is quadratic residue $x = a^{\frac{p-1}{2}}$

if $x \equiv 1 \pmod p$ then a is quadratic residue.

if $x \equiv -1 \pmod p$ then a is not quadratic residue

$\frac{p-1}{2}$ can be done in $O(\log p)$ multiplications so this alg. is efficient as p grows large

(2) g is a generator if & only if $g^{\frac{p-1}{q}} \neq 1 \pmod p$ for every prime factor q of $(p-1)$.
order of g is $p-1$

given p, $g \in \mathbb{Z}_p^*$

let $p-1 = \prod_{i=1}^{k} q_i^{e_i}$ where $q_1, q_2, \ldots, q_k$ are distinct prime factors of $p-1$

$\forall q \in q_1, \ldots, q_k$ find $h = g^{\frac{p-1}{q_i}} \bmod p$

if $h \equiv 1 \pmod p$, g is not a generator
if no h is $1 \bmod p$, then g is a generator

The exponentiation $g^{\frac{(p-1)}{q_i}} \bmod p$ can be done in $O(\log p)$ since we are given the factorization $p-1$ so the alg. is efficient.